

HERMAN REMINGTON

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [PeerList](#) | [Youtube](#)

SOFTWARE ENGINEER | FULL-STACK | SAAS | BACKEND | ENTERPRISE SYSTEMS

PROFESSIONAL SUMMARY

Software Engineer specializing in **SaaS platforms, enterprise systems, backend APIs, and full-stack application development** using Python, React, PostgreSQL, and modern cloud-native tooling. Demonstrated progression from building full-stack applications to designing **multi-tenant SaaS platforms, event-driven systems, financial workflows, AI integrations, and enterprise process automation solutions**.

Experienced in translating real-world operational requirements into scalable digital systems involving **role-based access control, organizational data isolation, workflow automation, payments, reporting, asynchronous processing, API integrations, observability, and containerized infrastructure**. Comfortable working across the full software development lifecycle, including requirements analysis, system architecture, backend development, database design, frontend development, testing, deployment, troubleshooting, and production maintenance.

Currently contributing to **ERP and Business Process Reengineering initiatives** within a large public-sector organization, building on previous experience developing SaaS products for property management, SMEs, payroll, AI automation, notifications, and healthcare.

CORE TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, HTML, CSS, SQL

Frontend: React.js, Bootstrap, Responsive UI, Modular CSS, Mobile-First Development

Backend & Frameworks: FastAPI, Flask, Laravel, SQLAlchemy, Frappe Framework, REST APIs

Databases: PostgreSQL, MySQL, MariaDB, SQLite, MongoDB

Architecture: SaaS Architecture, Multi-Tenant Systems, RBAC, Event-Driven Architecture, Microservice-Oriented Design, Workflow Automation, API Integration

Authentication & Security: JWT, OAuth, API Key Authentication, Role-Based Access Control, Session Management

Asynchronous Processing: Redis, Celery, Background Jobs, Task Queues

DevOps & Infrastructure: Docker, Docker Compose, Linux, Ubuntu, Nginx, Caddy, Git, CI/CD Workflows, Reverse Proxies, HTTPS, Persistent Storage

Cloud & Deployment: Vercel, Render, Supabase, HostAfrica, Ubuntu VPS

Monitoring & Reliability: Prometheus, Health Checks, Structured Logging, Request IDs, Service Monitoring

Payments & Integrations: M-Pesa, Stripe, OpenRouter API, Twilio, Deepgram

Enterprise Systems: ERPNext, Frappe Framework, MariaDB, Business Process Reengineering, Workflow Automation

Other: Postman, GitHub, SEO, Schema.org, Technical Documentation, UI/UX with Figma, Agile/Scrum

PROFESSIONAL EXPERIENCE

ICT Innovation & ERP Developer Intern

Kenya Broadcasting Corporation – ICT Department Innovation Team

April 2026 – April 2027 (Expected)

- Support **Business Process Reengineering (BPR)** initiatives across multiple organizational departments, analyzing existing manual processes and translating operational requirements into structured digital workflows.
- Contribute to the digitization of manual operational processes using **ERPNext and the Frappe Framework**, improving process visibility, accountability, and workflow consistency.
- Assist with the design, configuration, customization, testing, and implementation of **ERP modules and approval workflows** aligned with organizational requirements.
- Support automation of document handling, approvals, reporting, and inter-departmental operational procedures.
- Collaborate with technical and non-technical stakeholders to understand existing processes and translate them into practical enterprise system workflows.
- Participate in ERP configuration, testing, user support, troubleshooting, and iterative process improvements during implementation.
- Gain practical experience applying software engineering principles within a **large-scale public-sector enterprise environment**.

Technologies: ERPNext, Frappe Framework, Python, JavaScript, MariaDB, Linux, Git, Business Process Reengineering

Full-Stack Developer

Evolve Payroll & HR — Remote

[Github](#) | [Live site](#) / [Collaboration](#)

October 2025 – January 2026

Contributed to the development of a cloud-based **Payroll and HR SaaS platform** designed to support SMEs managing up to 500 employees.

- Developed payroll processing functionality covering Kenyan statutory deductions including **PAYE, NSSF, SHIF, Housing Levy, allowances, and salary adjustments**.
- Built employee management and **role-based access control** functionality to organize HR operations and protect business data.
- Implemented salary advance request and repayment workflows integrated with payroll calculations and payslip generation.
- Contributed to **M-Pesa and bank transfer payment workflows** for salary disbursement and reconciliation tracking.
- Developed secure REST APIs and responsive dashboards using **Laravel and React**.
- Worked within a multi-developer Agile environment using GitHub, contributing to sprint delivery, code reviews, debugging, and feature development.

Technologies: Laravel, React, PostgreSQL, JWT, Flutter, Git

Freelance Full-Stack Developer

BRS Holding — Remote

[GitHub](#)

May 2025 – June 2025

Customized and deployed an AI-powered content SaaS platform with subscription billing and usage-based monetization.

- Integrated **Claude through the OpenRouter API**, replacing the platform's default AI provider.
- Designed and implemented a **usage-based credit system** with daily quotas and feature restrictions to support SaaS monetization.
- Integrated **Stripe subscription billing** with Free, Pro, and Team pricing tiers.
- Implemented automated usage tracking and feature access based on subscription plans and available credits.
- Configured secure authentication and session management using **Supabase and Stack Auth**.
- Deployed and configured the application on an **Ubuntu Linux server**, managing environment configuration and production deployment workflows.

Technologies: React, Node.js ecosystem, Supabase, Stripe, OpenRouter API, Linux

Full-Stack Developer Intern

Moringa School

June 2024 – December 2024

- Developed multiple full-stack applications using **Flask, React, REST APIs, and relational databases**.
- Designed backend APIs and database schemas for application features and business workflows.
- Improved application responsiveness through database schema optimization and more efficient API integration.
- Collaborated on team-based software projects using Git and Agile development practices.
- Participated in peer code reviews, debugging, testing, and iterative feature development.
- Built foundational experience across frontend development, backend engineering, database design, API integration, and collaborative software delivery.

SELECTED ENGINEERING PROJECTS

Rental Property Management SaaS – AlphaOne

2026 – Present

[GitHub](#) | [Live Site](#) | [Collaboration](#)

Production-oriented multi-tenant property management platform designed for landlords, property managers, finance teams, tenants, and system administrators.

- Designed a **multi-tenant SaaS platform** supporting organizational data isolation and role-specific access across Landlords, Property Managers, Finance users, Tenants, and System Administrators.
- Designed and developed modular **REST APIs** supporting properties, units, leases, tenants, payments, expenses, inspections, maintenance tickets, vendors, reports, notifications, and financial operations.
- Implemented granular **Role-Based Access Control (RBAC)** across backend APIs and frontend application routes.
- Developed complete rental workflows covering property management, unit management, tenant onboarding, lease administration, rent collection, payment reconciliation, expenses, inspections, maintenance, and financial reporting.
- Designed database models and relationships using **SQLAlchemy and PostgreSQL** to support complex property and financial operations.
- Implemented **Docker and Docker Compose** infrastructure orchestrating PostgreSQL, Redis, backend, frontend, Caddy, and automated backup services.
- Configured Ubuntu Linux production infrastructure including reverse proxying, HTTPS, persistent database volumes, container orchestration, and automated PostgreSQL backups.

- Developed a separate public-facing marketing website with isolated CSS architecture, responsive navigation, SEO metadata, structured data, sitemap, robots configuration, and search-engine-friendly routes.
- Implemented technical SEO including **canonical URLs, Open Graph metadata, Twitter cards, Schema.org structured data, sitemap.xml, and robots.txt.**
- Established controlled Git branching and deployment workflows for application, SEO, and public website changes, reducing the risk of regressions during production releases.
- Built the frontend using **React and modular CSS**, maintaining separation between public-facing pages and authenticated SaaS application interfaces.

Technologies: Python, FastAPI, Flask, React, PostgreSQL, SQLAlchemy, Redis, Docker, Docker Compose, Caddy, Linux, Git, REST APIs, JWT, RBAC

Centralized Notification Platform – Notification Pipeline System

2026 – Present

[GitHub](#) | Demo

Designed and developed a centralized notification infrastructure platform for receiving application events and delivering notifications through multiple communication providers.

- Architected a modular notification platform using **FastAPI, PostgreSQL, Redis, Celery, and Docker**, separating API requests, asynchronous workers, persistence, and provider integrations.
- Developed REST APIs for **applications, API keys, events, notifications, templates, providers, monitoring, and reporting.**
- Implemented an **event-driven asynchronous processing architecture** using Redis and Celery, allowing notification workloads to execute independently from API requests.
- Designed a provider abstraction layer supporting multiple communication channels including **email, SMS, SMTP, WhatsApp, and Resend.**
- Implemented **API-key authentication, request identification, authentication middleware, structured logging, and protected API routes** for service-to-service communication.
- Developed notification lifecycle tracking covering delivery attempts, delivery logs, retries, provider resolution, and reporting.
- Implemented database versioning and schema evolution using **SQLAlchemy and Alembic.**
- Containerized the platform using Docker Compose with PostgreSQL, Redis, FastAPI, Celery workers, frontend, and Nginx services.
- Implemented production-oriented infrastructure on Ubuntu Linux including persistent storage, health checks, reverse proxying, service orchestration, and containerized frontend delivery.
- Added observability through **Prometheus metrics, health endpoints, structured request logging, and request ID middleware.**

- Diagnosed and resolved infrastructure issues involving Docker networking, service health checks, application startup, configuration, Git synchronization, and internal/external network routing.
- Established Git-based deployment workflows for synchronizing application changes with the production server and rebuilding containerized services.

Technologies: Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Redis, Celery, Docker, Docker Compose, Nginx, Linux, Prometheus, Git, REST APIs, API Authentication

SME Management SaaS Platform

2026 – Present

[GitHub](#) | [Live Site](#) | [Demo](#) | [Collaboration](#)

Multi-tenant SaaS platform designed to help small and medium-sized businesses manage sales, inventory, customers, credit, and business reporting.

- Designed **organization-scoped multi-tenant architecture** allowing multiple businesses to securely operate within a shared application.
- Developed sales processing workflows with automatic inventory deduction, customer credit tracking, debtor management, and staff performance attribution.
- Built backend APIs using **Flask and SQLAlchemy** supporting sales, inventory, customers, reporting, and operational workflows.
- Implemented asynchronous processing using **Celery and Redis** for background operations and resource-intensive tasks.
- Designed architecture for integration with the **Kenya Revenue Authority eTIMS API** for electronic tax invoice transmission.
- Developed mobile-first React interfaces optimized for frequent point-of-sale and inventory management workflows.
- Built reporting functionality covering sales summaries, profit estimates, inventory status, and business performance insights.
- Implemented authentication and organization-level authorization to protect business data.

Technologies: Python, Flask, React, PostgreSQL, SQLAlchemy, Celery, Redis, JWT, Vercel, Render

AI Real Estate Voice Agent

2025 – 2026

[GitHub](#) | [Demo](#)

- Built a real-time AI voice assistant for automating property inquiries and conversational workflows.
- Integrated **Twilio WebSockets** for real-time audio streaming and **Deepgram** for speech-to-text processing.

- Developed a WebSocket server capable of processing live audio streams and triggering domain-specific function calls.
- Designed modular integration architecture allowing additional property automation workflows to be added independently.

Technologies: Python, WebSockets, Twilio, Deepgram, Ngrok

Hospital Management System

2024 – 2025

[GitHub](#) | [Live Site](#) | [Figma Prototype](#)

- Developed a full-stack healthcare management system for patients, appointments, billing, inventory, and operational reporting.
- Implemented role-based authentication and protected access to application functionality and records.
- Built analytics dashboards for revenue and operational monitoring.
- Designed REST APIs and database workflows using Flask and PostgreSQL.

Technologies: Flask, React, PostgreSQL, Supabase

ENGINEERING GROWTH & HIGHLIGHTS

- Progressed from developing individual full-stack applications to designing **multi-tenant SaaS and enterprise platforms** with complex organizational, financial, and operational workflows.
- Designed and implemented systems spanning **property management, payroll, SME operations, notification infrastructure, AI automation, healthcare, and ERP process automation**.
- Developed practical expertise in **backend architecture, REST API design, relational database modeling, authentication, RBAC, asynchronous processing, third-party integrations, and frontend engineering**.
- Built event-driven architectures using **Redis and Celery**, separating background workloads from synchronous API operations.
- Developed provider abstraction patterns allowing external communication providers to be added without tightly coupling them to core application logic.
- Gained hands-on experience deploying and maintaining containerized systems using **Docker, Docker Compose, Linux, PostgreSQL, Redis, Nginx, and Caddy**.
- Implemented production-oriented observability using **Prometheus metrics, health checks, structured logging, and request ID tracing**.
- Developed experience troubleshooting real deployment issues involving **Docker networking, database connectivity, service health, configuration, Linux networking, reverse proxies, Git workflows, and container orchestration**.
- Applied technical SEO principles to production web applications including **structured data, canonical URLs, Open Graph metadata, sitemap generation, robots configuration, and search-friendly public routes**.

- Progressed from application development into **enterprise process analysis and Business Process Reengineering**, applying software engineering to real organizational workflows using ERPNext and Frappe.
- Demonstrated end-to-end ownership across requirements analysis, system architecture, backend development, database design, frontend implementation, integration, testing, deployment, monitoring, and production troubleshooting.

EDUCATION

Moringa School

Full-Stack Software Engineering Program

June 2024 – December 2024

ADDITIONAL INFORMATION

Development Practices: Agile/Scrum, Git-based workflows, Code Reviews, API Testing, Database Migrations, Technical Documentation

Engineering Focus: SaaS Development, Backend Engineering, Full-Stack Development, Enterprise Systems, ERP, Automation, Cloud-Native Applications, API Architecture